

Problem 5

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Problem 1 (a) If f is an odd function, why is it true that $\int_{-a}^a f(x) dx = 0$?

Support your reasoning with a picture.

Explanation. If f is odd, then the regions between the graph of f and the x -axis from $[-a, 0]$ and $[0, a]$ are reflections of each other through the origin. Thus, these two regions will have the same area but with opposite signs since they are on opposite sides of the x -axis. They will therefore cancel each other out.

(b) If f is an even function, why is it true that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$?

Support your reasoning with a picture.

Explanation. If f is even, then the regions between the graph of f and the x -axis from $[-a, 0]$ and $[0, a]$ are reflections of each other through the y -axis. Thus, these two regions will have the same area with the same sign since they are on the same sides of the x -axis. So you can only find one of these areas and then double it.